Symposium Report

Sex-based differences in endurance exercise muscle metabolism: impact on exercise and nutritional strategies to optimize health and performance in women

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New Findings

- **What is the topic of this review?**
  The topic is how sex influences carbohydrate and fat metabolism during exercise and whether this influences adaptation to nutritional and exercise regimens aiming to improve health and performance.
- **What advances does it highlight?**
  Women respond differently to certain nutritional and training regimens aimed at improving health and performance. Few studies have included women in trials and thus we are unsure how women respond to nutritional and training strategies aimed at improving health and performance.

Sex-based differences in substrate metabolism during moderate-intensity endurance exercise (END) have been well established. Specifically, during END of the same relative intensity women have a lower respiratory exchange ratio than men, indicative of a lesser reliance on carbohydrate oxidation to support fuel requirements for exercise. In fact, compared with men, women show a lesser reliance on both liver and muscle glycogen during END. Sex-based differences in intramyocellular lipid (IMCL) utilization during END are controversial. However, women have a larger depot of IMCL available to support END fuel needs and a greater percentage of IMCL in contact with mitochondria after a bout of END compared with men, suggestive of a greater capacity to use IMCL. These sex-based differences in metabolism during END are known to be mediated by oestrogen. Despite the well-recognized sexual dimorphisms in substrate metabolism during END, there is a paucity of research examining the effects of exercise and nutritional regimens aimed to enhance performance and/or health in women. Furthermore, the evidence that does exist is suggestive of discordance in the effectiveness of nutritional and exercise regimens between the sexes. The focus of this review is to provide an overview of the well-established sex-based differences in metabolism during END and how they relate to the physiological responses to nutritional and exercise strategies intended to improve exercise performance and/or health.

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Introduction

We are entering the era of personalized nutritional and training strategies for optimal health and performance and yet the majority of the research that supports any given strategy to enhance health or endurance exercise (END) performance has been conducted in men. Studies

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frequently exclude women as participants because of the potential influence of fluctuating ovarian hormones throughout the menstrual cycle and the potential impact this might have on the outcomes of interest. However, there are well-recognized sex-based differences in substrate metabolism during END (Tarnopolsky *et al.* 1990, 1995; Romijn *et al.* 2000; Carter *et al.* 2001*b*; Devries *et al.* 2006) that may influence how women respond to a given training/nutritional strategy. The purpose of this review is to provide an overview of the well-established sex-based differences in metabolism between men and premenopausal women during END, briefly discuss how these differences are mediated by oestrogen and highlight how these sex-based differences may impact adaptations to nutritional and exercise regimens intended to improve performance and/or health.

Sex-based differences in metabolism during exercise

Sex-based differences in substrate metabolism during moderate-intensity END between men and premenopausal women have been well established (Fig. 1). Specifically, women have a lower respiratory exchange ratio during END than men (Tarnopolsky *et al.* 1990, 1995; Romijn *et al.* 2000; Carter *et al.* 2001*b*; Devries *et al.* 2006), indicative of a lesser reliance on whole-body carbohydrate oxidation to support fuel requirements for END. At the organ level, it has been consistently shown that, compared with men, women have a lower reliance on liver carbohydrate stores [decreased glucose rate of appearance (R_a), rate of disappearance (R_d) and metabolic clearance rate; Friedlander *et al.* 1998; Carter *et al.* 2001*b*; Devries *et al.* 2006], although it has not been determined whether this is a result of lower liver glycogenolysis, gluconeogenesis or both. Whether muscle glycogen stores are spared during END in women is contentious. Some studies have found no effect of sex on muscle glycogen utilization during cycling END (Tarnopolsky *et al.* 1995; Roepstorff *et al.* 2002; Zehnder *et al.* 2005), whereas others have found, using a running (Tarnopolsky *et al.* 1990) as opposed to cycling, or a Wingate protocol (30 s high-intensity bike sprint; Esbjörnsson-Liljedahl *et al.* 1999), that women utilize 25% and 50% less muscle glycogen, respectively, during the exercise bout compared with men. Furthermore, we have previously shown that when women were tested in the luteal phase, but not the follicular phase, of the menstrual cycle, they used less proglycogen (the dynamic form of glycogen) during a 90 min bike ride at 65% of peak oxygen uptake ($\dot{V}_{O_2peak}$) compared with men (Devries *et al.* 2006). Thus, it appears as if, depending on the type of exercise and/or the phase of the menstrual cycle in which the exercise is performed, women may spare muscle glycogen during exercise.

Sex also influences lipid metabolism during END as evidenced by a higher glycerol R_a during exercise in women (Carter *et al.* 2001*b*); however, the source [plasma free fatty acids (FFA) and/or intramyocellular lipid (IMCL)] of the increased glycerol (reflecting lipolysis) during END remains unknown. At various exercise intensities (25, 45, 65 and 85% $\dot{V}_{O_2peak}$), no sex-based differences in FFA oxidation during exercise were observed (Burguera *et al.* 2000; Romijn *et al.* 2000). However, a single study by Roepstorff *et al.* (2002) examined participants who cycled for 90 min at 60% $\dot{V}_{O_2peak}$ and found that, although not statistically significant, during the last 60 min of the exercise bout women oxidized 47% more plasma FFA than men. These findings are difficult to interpret because during this time period there were no differences in whole-body substrate oxidation between sexes. Nevertheless, these findings do suggest that as the duration of the exercise bout progresses women may begin to rely to a greater extent on the use of plasma FFA for fuel.

Perhaps even more controversial than sex effects on plasma FFA oxidation is whether women rely to a greater extent on IMCL during END in comparison to men. Although it is well recognized that women have a greater IMCL content than men (Roepstorff *et al.* 2002; Devries *et al.* 2007; Tarnopolsky *et al.* 2007), studies examining sex-based differences in IMCL utilization during END have been equivocal, finding that women have a greater (Roepstorff *et al.* 2002, 2006; Steffensen *et al.* 2002), lesser (Zehnder *et al.* 2005) or equal IMCL utilization (White *et al.* 2003; Devries *et al.* 2007) during END. Limitations in the methodologies used and differences in training status between participants within a trial further complicate this issue. However, the findings that women have an increased IMCL content and that following END the percentage of IMCL in direct contact with mitochondria increases suggest that women have a greater capacity to use IMCL, and perhaps with increasing exercise duration, sex-based differences may be more apparent.

Lastly, sex also influences protein oxidation during exercise, particularly of the branched chain amino acid leucine. In comparison to men, premenopausal women oxidize less leucine during endurance exercise (Phillips *et al.* 1993; McKenzie *et al.* 2000; Lamont *et al.* 2001). Furthermore, non-oxidative leucine disposal (reflective of whole-body protein synthesis) is greater in women during endurance exercise than in men (Lamont *et al.* 2001). Taken together, the findings from sex comparative studies show that in the fasted state premenopausal women rely to a greater extent on fat sources and to a lesser extent on carbohydrate and protein sources during endurance exercise.

Sex-based differences in metabolism are mediated by oestrogen

Whether these sex-based differences in metabolism are mediated by differences in oestrogen between the sexes

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has been examined in several studies (Ruby *et al.* 1997; Carter *et al.* 2001*a*; Tarnopolsky *et al.* 2001*a*; Devries *et al.* 2005; Hamadeh *et al.* 2005). Short-term supplementation of oestrogen (doses to mimic follicular phase, luteal phase and supraphysiological oestrogen concentrations) administered to ammenorrhoeic women or to men decreased liver glucose output and muscle uptake (Ruby *et al.* 1997; Carter *et al.* 2001*a*; Devries *et al.* 2005), with no effect on whole-body lipolysis (measured as glycerol R_a and R_d; Ruby *et al.* 1997; Carter *et al.* 2001*a*) or muscle glycogen utilization (Tarnopolsky *et al.* 2001*a*; Devries *et al.* 2005). Additionally, oestrogen supplementation has been found to increase plasma FFA concentrations (Ruby *et al.* 1997) and lower muscle glycogen content (Devries *et al.* 2005). Furthermore, in the only oestrogen supplementation trial that measured leucine kinetics during endurance exercise (Hamadeh *et al.* 2005), after oestrogen treatment leucine oxidation during exercise was lower, with no change in protein synthesis or breakdown during exercise, which resulted overall in a less negative state of negative protein balance during exercise compared with placebo. Despite these shifts in substrate availability, only one study to date has found that oestrogen supplementation decreased the respiratory exchange ratio and whole-body carbohydrate oxidation (Devries *et al.* 2005). Differences in dosing parameters (length of supplementation, low *versus* high dose), study size and subject sex (ammenorrhoeic women *versus* men) may explain these differential findings. Collectively, these data suggest that oestrogen acts primarily on the liver, decreasing glucose release to influence substrate metabolism during END.

Importantly, given the opposing action of oestrogen supplementation on circulating testosterone concentrations, whether the effects of oestrogen are the result of increased circulating oestrogen or decreased circulating testosterone is often queried. However, neither elevation of circulating testosterone concentrations ($\sim$10.0 ng ml^{-1}) using transdermal testosterone patches (10 mg day^{-1} for 3 days) nor suppression of circulating testosterone concentration ($\sim$0.5 ng ml^{-1}) using a gonadotrophin-releasing hormone antagonist (3 mg day^{-1} for 3 days, s.c. injection) altered liver glucose release, estimated muscle glycogen utilization or whole-body substrate utilization in young men during END (Braun *et al.* 2005). As such, it is likely to be the differences in oestrogen, and not testosterone, that are responsible for the sex-based differences in metabolism observed during END.

Another sex hormone that might potentially alter substrate metabolism during exercise in women is progesterone. A complete examination of how progesterone and oestrogen interact in regulation of substrate metabolism during exercise is beyond the scope of this review, and the reader is directed to a review by Oosthuyse & Bosch (2010) for more information.

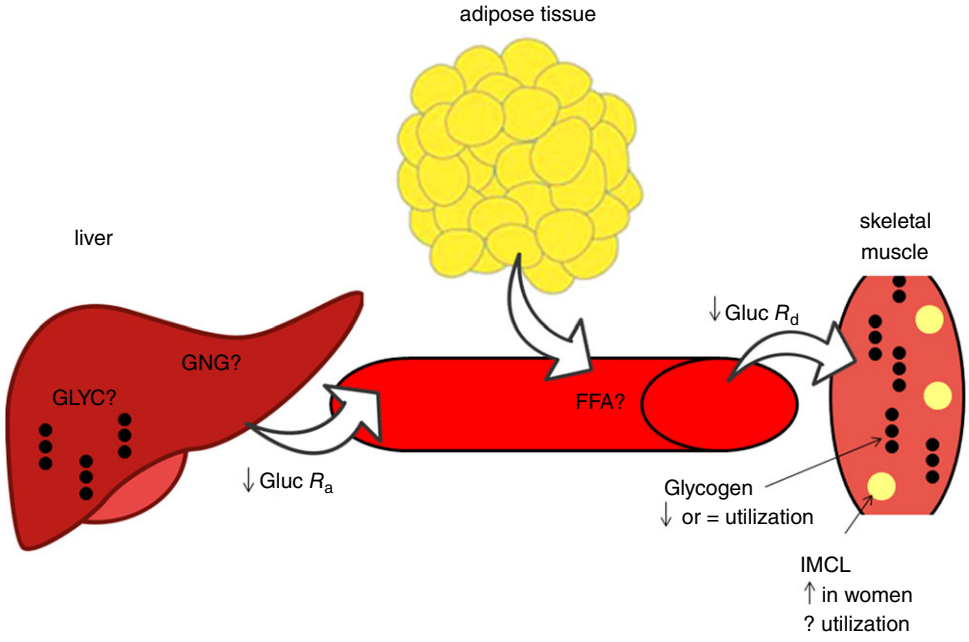


Figure 1. Schematic overview of the known and potential sites that underpin sex-based differences in fuel utilization during endurance exercise
Abbreviations: FFA, free fatty acids; Gluc, glucose; GLYC, glycogenolysis; GNG, gluconeogenesis; IMCL, intramyocellular lipid; R_a, rate of appearance; and R_d, rate of disappearance.

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Briefly, progesterone is considered to act in an opposing manner to oestrogen; however, its role in influencing substrate utilization during exercise has not been fully established. While several studies examining the role of oestrogen supplementation have been conducted (see above), no studies have examined the role of progesterone alone on endurance exercise substrate selection. One trial has compared oestrogen supplementation with oestrogen and progesterone supplementation (E+P) and ovarian hormone suppression (D'Eon *et al.* 2002) and found that total carbohydrate oxidation and estimated muscle glycogen utilization during endurance exercise were lower following oestrogen treatment than with E+P and hormone suppression treatments. On the contrary, glucose R_a during exercise was lower following both oestrogen and E+P treatments compared with ovarian hormone suppression. These findings suggest that progesterone acts on the muscle, but not the liver, to prevent glycogen sparing. However, these findings conflict with those of studies comparing women in different phases of the menstrual cycle that have shown that muscle glycogen utilization is lower in the luteal phase, when oestrogen and progesterone are both high but progesterone predominates, compared with the follicular phase, when oestrogen and progesterone are both low but oestrogen predominates (Hackney, 1999; Devries *et al.* 2006). As such, the role for progesterone in the regulation of fuel utilization during endurance exercise remains unclear and should be examined.

Could sex-based differences in metabolism during exercise impact the response to training and dietary strategies aimed at improving health and performance?

Given the well-established sex-based differences in metabolism during END, one important question remains: do women respond differently to strategies aimed at improving health and performance? In short, we are unable to answer that question at this time. As mentioned above, most studies to date examining the effects of an intervention on health and/or performance have done so using men as participants. As such, very few data exist on how women respond to such interventions. Nonetheless, there are some data from studies using both men and women that would suggest that, at least in some situations, women respond differently in response to a training/nutritional regimen. The purpose of this section is to highlight some examples in which differences between men and women have been found. Importantly, this is not an exhaustive list of studies that have been conducted in both men and women; it is simply meant to illustrate that we cannot assume that women will respond in the same manner as men to a training/nutritional strategy and that it is important to determine the specific effects of these interventions in women.

As detailed above, women rely to a greater extent than men on lipid sources to fuel a bout of END (Tarnopolsky *et al.* 1990, 1995; Romijn *et al.* 2000; Carter *et al.* 2001*b*; Devries *et al.* 2006). Additionally, it is well understood that habitual dietary intake influences substrate utilization during exercise, with high-fat diets increasing fat oxidation (Burke *et al.* 2002; Stellingwerff *et al.* 2006) and high-carbohydrate diets increasing carbohydrate oxidation (Tarnopolsky *et al.* 1995). One strategy often used by athletes to improve performance is carbohydrate loading (Tarnopolsky *et al.* 1995). However, using a traditional carbohydrate-loading regimen (75% energy intake from carbohydrates), Tarnopolsky *et al.* (1995) found an increase in muscle glycogen stores and enhanced performance in men but not women. Further exploration revealed that when women consumed 75% of energy from carbohydrates their daily carbohydrate intake relative to fat-free mass (FFM) fell short [at 5.9–7.9 g (kg FFM)$^{-1}$ day^{-1}] of the 8.5 g (kg FFM)$^{-1}$ day^{-1} required to induce glycogen supercompensation (Tarnopolsky *et al.* 1995, 2001*b*). In fact, in order to increase muscle glycogen stores women had to increase their energy intake by 34% to reach this threshold (Tarnopolsky *et al.* 2001*b*). Furthermore, when women increased their carbohydrate intake to 10 g (kg FFM)$^{-1}$ day^{-1}, they were able to increase muscle glycogen stores and improve performance; however, the increase in muscle glycogen stores was less than that in men (Walker *et al.* 2000). Thus, increasing caloric intake or relative carbohydrate intake to supercompensate glycogen may not be an appropriate strategy for women given that athletes may be apprehensive to increase caloric intake in the days leading up to an event owing to the potential deleterious effects on performance.

Given the increased reliance on fat stores to support END in women, a performance advantage for women consuming a high-fat diet may exist. However, despite this knowledge, few studies examining the effects of fat loading or fat adaptation and carbohydrate restoration on performance have included women (reviewed by Yeo *et al.* 2011). Fat-adaptation diets involve consumption of a high-fat, low-carbohydrate diet and result in increased rates of fat oxidation and decreased muscle glycogenolysis during END, which persist upon carbohydrate restoration (Yeo *et al.* 2011). Although the findings from studies examining the impact of such strategies on performance are mixed (summarized by Yeo *et al.* 2011), given that women not only use more lipid during END at all intensities (Devries *et al.* 2006; Cheneviere *et al.* 2011) but also have greater IMCL available for oxidation (Devries *et al.* 2007) and can exercise at a higher intensity prior to

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reaching their maximal fat oxidation rate (58 *versus* 50% $\dot{V}_{O_2peak}$ in women and men, respectively; Cheneviere *et al.* 2011), this suggests that women may respond favourably to such an intervention. However, it is possible that women may already have maximally adapted their ability to oxidize lipids and thus fat-adaptation strategies may not be effective; nevertheless, the finding that the site of carbohydrate sparing induced by sex is predominantly the sparing of liver carbohydrate stores (Carter *et al.* 2001*b*; Devries *et al.* 2006), whereas fat adaptation decreases muscle glycogen utilization, suggests differential sites of regulation and potentially synergistic effects on fat oxidation. In summary, fuel utilization during exercise is influenced by dietary intake in the days leading up to and during the exercise bout, and given the inherent differences in substrate metabolism during END in the fasted state, the preferential fuel to enhance performance in women may be different from that in men and should be explored.

It is not only sport performance where the benefits of exercise and nutrition in women are not fully understood; the effects of specific exercise and nutritional regimens to optimize health in women are also relatively unknown. For example, while high-intensity interval training (HIIT; for the purpose of this review including both sprint- and high-intensity interval training) increases aerobic capacity in both men and women, its ability to enhance insulin sensitivity appears to be blunted in women (Metcalfe *et al.* 2012; Gillen *et al.* 2013). It is purported that high muscle glycogen utilization during HIIT and subsequent glycogen resynthesis induces these rapid changes in insulin sensitivity (Metcalfe *et al.* 2012). As noted above, during a single bout of HIIT women use 50% less muscle glycogen than men (Esbjörnsson-Liljedahl *et al.* 1999), which may underpin the attenuated (or absent) effect of HIIT on insulin sensitivity in women. High-intensity interval training is an attractive exercise strategy because of its low time commitment; however, based on preliminary, short-term studies, it may not be the ideal exercise strategy for women looking to enhance insulin sensitivity (i.e. prediabetics and diabetics). An understanding of the underlying mechanisms responsible for this differential response is needed to optimize diabetes prevention in both sexes, particularly with ageing, because older women appear to be at an increased risk for insulin resistance. Additionally, further research examining the long-term effects of HIIT on insulin sensitivity in women is warranted.

Given that women oxidize less leucine during endurance exercise (Phillips *et al.* 1993; McKenzie *et al.* 2000; Lamont *et al.* 2001), it would not be surprising if other sex-based differences in protein metabolism existed. Although much of the work examining the muscle protein synthetic response to protein feeding has involved male subjects, one study has examined and found no sex-based differences in anabolic sensitivity between young men and women (West *et al.* 2012). However, with respect to anabolic sensitivity there seems to be an interaction between age and sex whereby as both men and women get older they become less anabolically sensitive than their younger counterparts; however, in addition, older women become less anabolically sensitive than their older male counterparts (Smith *et al.* 2008). These findings have led to recommendations that the protein requirements for older adults be increased from the recommended daily allowance (RDA; 0.8 g kg^{-1} day^{-1}) to 1.0–1.5 g kg^{-1} day^{-1} (Wolfe *et al.* 2008; Bauer *et al.* 2013). In fact, work from Moore *et al.* (2015) found that compared with an optimal dose per meal of 0.24 g kg^{-1} in young men, a dose of 0.4 g kg^{-1} is required to optimize muscle protein synthesis in older men, which equals 1.2 g kg^{-1} day^{-1} when translated over three meals per day. However, the optimal dose per meal to support muscle protein synthesis in older women has not been determined. Clinically, this becomes relevant because although younger individuals consume protein above the RDA (1.3 g kg^{-1} day^{-1}; Fulgoni, 2008), 40% of older adults are not meeting the current protein RDA and 10% of older women are not even meeting the estimated average requirement for protein intake (0.66 g kg^{-1} day^{-1}; Houston *et al.* 2008). Given that older women are less anabolically sensitive than their age-matched male counterparts (Smith *et al.* 2008), the optimal protein dose per meal in older women is likely to be greater than that for older men. Given the progressive loss of muscle mass and strength that accompanies ageing and its association with increased risk of falls, metabolic disease and decreased quality of life, knowing the appropriate dose of protein to study (and eventually to recommend to older women) is of utmost importance to optimize the muscle and overall health of older women and is yet another example of how sex may influence the response to health strategies.

In summary, there are well-known sex-based differences in fuel utilization during END. These metabolic differences may impact how women respond to nutritional and exercise regimens aimed at improving health and/or performance. To date, very little research has been conducted examining whether women respond in a similar manner to men in response to such interventions. Although much of the discussion above pertaining to how women may respond to performance and health strategies is speculative, it is because in many areas there is a paucity of research examining such strategies in women. In the future, research needs to be conducted using women as subjects to determine the impact of, and mechanisms that underpin, nutritional and exercise interventions in order to optimize the health and performance of women.

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Additional information

Competing interests

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